EVADING FIREWALLS LAB REPORT

BACKGROUND	
Course:	CIS 170 - Hands-On Ethical Hacking and Network Defense
Term:	Spring 2026
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Submission Platform:	Canvas
Report Date:	2026-03-05
Report Version:	1.0
Lab Title:	Evading Firewalls
Module Reference:	Evading Firewalls
Lab Type:	Live Virtual Machine Lab
Date Completed:	2026-03-05
Time Spent:	Approximately 1 hour

ENVIRONMENT

Platforms and Systems:

- Primary environment: Practice Labs / Live Virtual Machine Labs controlled sandbox.
- Primary systems available in the topology: PLABDC01 (Windows Server 2019 - Domain Controller), PLABDM01 (Windows Server 2019 - Domain Member), PLABWIN10 (Windows 10 - Domain Member), and PLABKALI01 (Kali 2019.2 - Linux Kali Workstation).
- The work centered mainly on PLABWIN10 using Internet Explorer for AVG Firewall installation, scans, logs, and anonymous proxy testing with proxfree.com.
- Remote access to the virtual machines was provided through the course platform.

Context and Constraints:

- The module focused on installing/configuring AVG Firewall and bypassing blocked sites using anonymous proxy websites (proxfree.com).
- Practice Labs flagged the module as legacy content using deprecated software, so behavior could vary and some steps were best interpreted as guided sandbox demonstrations.
- Windows Updates were disabled in the Practice Labs environment, so software behavior reflected a controlled legacy setup rather than a current hardened enterprise baseline.
- The module aligned most closely with exam objectives 4.3 (Information Security Tools).

Authorization and Scope:

- All documented activity took place only inside approved academic systems and isolated practice-lab sandboxes.
- No public IP space, real organizations, or unauthorized targets were involved.
- No production infrastructure, third-party systems, or live services were contacted.
- The actions documented here were performed for defensive understanding, firewall evasion, and ethical hacking education within the course environment.
- Restricted material, live credentials, and personal data are omitted or redacted from this report.

Work Objectives:

- Install and configure AVG Free Firewall on PLABWIN10.
- Perform AVG Smart Scan and review logs.
- Use anonymous proxy site (proxfree.com) to bypass blocked websites.

Evading Firewalls Concepts and Defensive Framing:

- The lab identified two core phases of firewall evasion: installing/configuring a personal firewall (AVG) and bypassing corporate blocks using anonymous proxy websites.
- Techniques reviewed included AVG Smart Scan, log review, and proxy-based web surfing.
- Tools demonstrated both commercial (AVG Firewall) and web-based (proxfree.com) approaches, showing how firewalls can be configured or bypassed.
- Prevention concepts reviewed included proper firewall rules, monitoring logs, and restricting anonymous proxy usage, reinforcing that firewall defense requires both technical controls and policy enforcement.

TECHNICAL ANALYSIS
AVG Firewall Installation and Scanning on PLABWIN10:
– On PLABWIN10, Internet Explorer was used to navigate to https://www.avg.com/en-us/free-antivirus-download. – AVG Free Antivirus setup was downloaded and installed (avg_antivirus_free_setup.exe). – During installation, "Yes, Install AVG Secure Browser" was unticked. – After installation, AVG launched and the user selected CONTINUE WITH FREE. – AVG Smart Scan was executed (RUN SMART SCAN), scanning the operating system for threats. – The scan completed with two issues found, but they were skipped for this lab. – Logs were accessed via Web & Email → Open (Enhanced Firewall) → Logs to review application activity.
Anonymous Proxy Usage with proxfree.com:
– Internet Explorer was used to visit http://www.hongkiat.com/blog/how-to-access-blocked-web-sites/ to review a list of anonymous proxy sites. – The proxy site https://www.proxfree.com was accessed (allowed in the lab environment). – The URL www.google.co.uk was entered in the proxfree textbox and PROXFREE was clicked. – The Google UK homepage loaded through the proxy, with the address bar showing the proxied connection. – This successfully demonstrated bypassing potential firewall blocks using an anonymous proxy.

KEY COMMANDS, TOOLS, AND ARTIFACTS BY SYSTEM
PLABWIN10 — AVG Firewall & Proxy Tools:

`https://www.avg.com/en-us/free-antivirus-download (AVG Free download)`
`AVG Smart Scan executed via GUI`
`Web & Email → Open (Enhanced Firewall) → Logs (review application logs)`
 	
`https://www.proxfree.com (anonymous proxy)`
`www.google.co.uk entered in proxfree textbox → PROXFREE clicked`

Primary Tools and Artifacts Observed:
– AVG Free Firewall successfully installed and Smart Scan completed with no malware detected. – AVG logs reviewed showing application activity and firewall events. – proxfree.com successfully proxied www.google.co.uk, demonstrating bypass of potential firewall blocks. – The most important evidence of successful firewall evasion in the lab was the proxied Google page loading through proxfree.com and the completed AVG Smart Scan/logs.

DEFENSIVE MAPPING AND SUMMARY

Technique Summary:

- The completed work covered AVG Firewall installation, Smart Scan, log review, and anonymous proxy usage (proxfree.com) to bypass blocked sites.
- The recurring theme across the module was that firewalls can be evaded using personal firewall software and anonymous proxies.

MITRE ATT&CK Mapping:

- Defense Evasion: T1562.004 Disable or Modify System Firewall - AVG Firewall configuration.
- Command and Control: T1090.002 Proxy through Compromised Infrastructure - anonymous proxy usage.
- Discovery: T1595.002 Vulnerability Scanning - AVG Smart Scan for threats.

Detection Opportunities:

- Unexpected firewall software installations (AVG), unusual proxy traffic, and anonymous proxy site access are strong indicators of evasion.
- Endpoint monitoring for AVG installation and proxy domains provides visibility.
- Web proxy logs and firewall logs should be monitored for anomalous outbound connections.

Defensive Controls:

- Implement strict application whitelisting to prevent unauthorized firewall/proxy software installation.
- Block known anonymous proxy domains at the network level and monitor outbound web traffic.
- Enable centralized logging and regular firewall/endpoint audits.
- Use next-generation firewalls with application awareness to detect proxy usage.

Completion Verification:

- Work was completed across AVG Firewall installation, Smart Scan, log review, and anonymous proxy testing.
- Completion was evidenced by successful AVG installation/scan, log access, and proxied Google page.
- The actual steps and tool outputs used in the lab are preserved directly in this report for accurate documentation and reproducibility.

Key Findings:

1) AVG Firewall provides personal firewall protection with Smart Scan and logging capabilities.
2) Anonymous proxies (proxfree.com) can effectively bypass web filtering and blocked sites.
3) Logs in AVG reveal application activity and firewall events for monitoring.
4) Organizations must monitor for unauthorized proxy usage and firewall software installations.
5) The strongest defensive takeaway was that layered controls (application whitelisting, proxy blocking, and logging) are essential to prevent firewall evasion.

Issues / Deviations / Notes:

- The lab used deprecated software and a legacy environment, so exact tool behavior could vary.
- All activities were performed only inside the isolated Practice Labs sandbox.

Reflection:

- This lab tied together personal firewall installation/configuration and anonymous proxy usage into a complete firewall evasion workflow.
- The strongest takeaway was that firewalls can be evaded using both local software and web proxies, making strict application control and proxy monitoring essential for defense.